

$^{13}\text{N } \varepsilon + \beta^+ \text{ decay}$

Type	Author	History	Citation	Literature Cutoff Date
Full Evaluation	J. H. Kelley, C. G. Sheu and J. E. Purcell		NDS 198,1 (2024)	1-Aug-2024

Parent: ^{13}N : $E=0$; $J^\pi=1/2^-$; $T_{1/2}=9.9584 \text{ min } 36$; $Q(\varepsilon)=2220.47 \text{ 27}$; $\% \varepsilon + \% \beta^+ \text{ decay}=100$

^{13}N - $T_{1/2}$: From Adopted Levels of ^{13}N .

^{13}N - $Q(\varepsilon)$: From (2021Wa16).

1935Ru01: $^{13}\text{N}(\beta^+)$; $^{10}\text{B}(\alpha, n)$ was used to produce ^{13}N , which was found to decay with $T_{1/2} \approx 14$ minutes.

1939Wa09: measured $T_{1/2}=9.93 \text{ min } 3$.

1948Co24: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=10.2 \text{ min } 1$.

1950Ho01: $^{13}\text{N}(\beta^+)$; measured end-point energy $E_\beta(\text{max})=1.202 \text{ MeV } 5$ and $T_{1/2}=10.05 \text{ min } 10$.

1953Ch34: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=602.9 \text{ sec } 19=10.05 \text{ min } 3$.

1954Gr66: $^{13}\text{N}(\beta^+)$; measured $E_\beta(\text{max})=1.185 \text{ MeV } 25$.

1955Wi43: ^{13}N ; measured $T_{1/2}=10.08 \text{ min } 4$.

1957Da08: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=9.96 \text{ min } 3$.

1957De22: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=10.02 \text{ min } 10$.

1957No17: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=10.07 \text{ min } 6$.

1958Ar15: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=9.96 \text{ min } 3$.

1958Da09: $^{13}\text{N}(\beta^+)$; measured $E_\beta(\text{max})=1.190 \text{ MeV } 3$ and $T_{1/2}$ as in (1957Da08).

1960Ja12: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=9.9965 \text{ min } 5$.

1960Ki02: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=9.93 \text{ min } 5$.

1961Ra06: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=12.3 \text{ min } 7$ (excluded).

1965Bo42: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=10.05 \text{ min } 5$.

1965Eb01: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=9.96 \text{ min } 2$.

1968Ri15: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=9.963 \text{ min } 9$.

1971Go40: $^{13}\text{N}(\beta^+)$; for decay to $^{13}\text{C}_{\text{g.s.}}$, $\log ft=3.667 \text{ 1}$.

1973SiYS: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=10.0 \text{ min } 5$.

1977Az01: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=9.965 \text{ min } 10$.

1980An40: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=9.967 \text{ min } 10$.

1989KaYR: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=9.962 \text{ min } 20$.

2022Lo14: $^{13}\text{N}(\beta^+)$; measured $T_{1/2}=9.9508 \text{ min } 32$; provided an analysis using only their value and (1968Ri15, 1977Az01) and recommended $T_{1/2}=9.9532 \text{ min } 37$.

1995Va27: $^{13}\text{N}(\beta^+)$; measured β asymmetry following beam implantation in Pt; deduced beam polarization.

Theory:

1970Ko41: general model for β decay in even-odd nuclei.

1970Da21: generalized pairing-force model analysis of $\log ft$ values.

1972Ma72: calculated β shape spectra.

1973Wi04: analysis of the Axial-Vector coupling constant.

1973Su04: analysis of K-electron capture branching ratios.

1975Kr14: developed O(5) symmetry model to analyzed $\log ft$ values.

1977Ri08: Shell model analysis of $\log ft$ values.

1980An31: analysis of K-electron capture rates.

1984Ko40: analyzed isotensor component in isovectorial transition.

1991Na05: analyzed mirror nuclei decays in a search for evidence of right-handed currents.

1995Go34: analyzed β -decay polarization asymmetry data in a search for evidence of right-handed chirality.

2008Se10: analyzed half-lives, branching ratios, electron-capture probabilities; deduced ft values in mirror decays.

2008Pe13: analyzed correlations between β decays in mirror nuclei and their magnetic moments.

2012Sa50: analysis of isospin related corrections for superallowed β transitions.

2015Mo10, 2010MoZU: calculated improved beta spectra, shape factors, mean energies, experimental mean energies.

2015To02: developed parametrization of the statistical rate functions, f , for superallowed $T=1/2$ transitions.

2021Ir01: single-particle model analysis of the $^{13}\text{N}(\beta^+)$ reaction. Emphasized asymptotic normalization coefficients (ANCs) and

¹³N ε+β⁺ decay (continued)

spectroscopic factor data.
[2021Da11](#): analyzed the role of tensor forces in β decay.
[2024Fa01](#): analyzed decay between isobaric analog mirror states.

¹³C Levels

<u>E(level)</u>	<u>J^π</u>	<u>T_{1/2}</u>
0	1/2 ⁻	stable

ε,β⁺ radiations

εK, εL, εM, εN: [Additional information 1](#).
av Eβ: [Additional information 2](#).

<u>E(decay)</u>	<u>E(level)</u>	<u>Iβ⁺ [†]</u>	<u>Iε [†]</u>	<u>Log ft</u>	<u>I(ε+β⁺) [†]</u>	<u>Comments</u>
(2220.5 15)	0	99.803 5	0.197 5	3.67174 44	100	av Eβ=492.21 12; εK=0.00182 5; εL=1.466×10 ⁻⁴ 42 C1=0.001 24 (1958Da09). Cexp(W) = 1+C ₁ W.

[†] Absolute intensity per 100 decays.